

Scenario analysis of electricity pathways in Kenya using Low Emissions Analysis Platform and the Next Energy Modeling system for optimization

Valentine Wambui^{a,*}, Francis Njoka^b, Joseph Muguthu^b, Patrick Ndwali^c

^a Kenya University, P. O. Box 43844 – 00100, Kenya

^b Department of Energy Technology, Kenya University, Kenya

^c College of Engineering and Technology, Jomo Kenyatta University of Agriculture and Technology, P. O. Box 62000 – 00200, Kenya

ARTICLE INFO

Keywords:

Energy modeling
LEAP-NEMO
Energy optimization
High RE penetration
Environmental externalities

ABSTRACT

Many nations have encouraged the need to fully exploit renewable energy resources and energy-efficient technologies to address the issue of high electricity demand and reduce greenhouse gas emissions. The development of energy pathways and sustainable energy policies of diverse countries for long-term planning has always been a challenging task. This paper investigates the impact of alternative electricity development pathways on system costs and greenhouse gas emissions in Kenya. The Low Emissions Analysis Platform (LEAP) with Next Energy Modeling system for Optimization (NEMO) is used to evaluate the optimized least-cost power system and energy storage. Three scenarios are analyzed: high penetration of renewable energy resources with an energy storage system, internalizing of environmental externalities, and energy productivity. Results from the study indicate that with Kenya's Least Cost Power Development Plan 2017–2037 as the reference, the overall cost savings are up to 26.1 billion US dollars which is equivalent to 55% while the carbon emission saved with high penetration of renewable energy with storage is up to 6.3 Mt. On the other hand, energy productivity shows a downward trend with the existing policies and exhibits an upward trend when two policy instruments are considered. Consequently, those alternative energy planning pathways have the potential to significantly reduce overall system costs and GHG emissions.

1. Introduction

Electrical energy is a crucial element of socio-economic development for any country [1]. Undoubtedly, conventional fossil fuels are presently the main sources of electricity generation, however, they produce large amounts of greenhouse gas (GHG) emissions. These GHGs contribute to climate change which is currently a major global environmental risk [2]. Furthermore, two-thirds of Africans, equivalent to 675 million people, do not have access to electricity. The per capita electricity consumption is very low in the majority of Sub-Saharan African (SSA) countries. This lack of electricity prevents the socio-economic development of SSA countries [3]. Kenya, which is one of the SSA countries, seeks to attain middle-income status by 2030. Industrialization has been identified as a key pillar to achieving the vision 2030 outcomes, and this requires growth in energy supply. Historically, industrial revolutions are largely driven by fossil fuels significantly increasing the concentration of the GHGs. Providing abundant energy without causing deterioration of the environment is always an immense challenge [4].

Kenya through the National Adaptation Plan (NAP) 2015–2030, demonstrates her commitments to the Paris Agreement and puts forth critical responses to the impacts of climate change. Through the National Climate Change Action Plan (NCCAP) 2018–2022, the country further commits to pursuing climate-proof energy and transport infrastructure; encouraging electricity supply based on renewable energy; encouraging the transition to clean cooking; and developing sustainable transport systems [5]. Although the Government of Kenya (GoK) has committed to lowering its GHG emissions with respect to PA, it has also planned for coal, nuclear, and natural gas in her Least Cost Power Development Plan (LCPDP) 2017–2037 to meet this anticipated demand. Despite Kenya's energy system being mainly composed of RERs namely, geothermal, hydropower, wind, and solar energy, among others, the country can experience a considerable increase in GHG emissions if coal and natural gas are added to the electricity generation mix.

Renewable energy resources (RERs) play a significant role due to their availability, cleanliness, and scalability [6]. However, most of the RERs are unreliable, variable, and present a stochastic nature. As a

* Corresponding author.

E-mail address: valentinewambui@students.ku.ac.ke (V. Wambui).

<https://doi.org/10.1016/j.rser.2022.112871>

Received 10 March 2022; Received in revised form 23 July 2022; Accepted 15 August 2022

Available online 30 August 2022

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Nomenclature and abbreviations			
BAU	Business as Usual	Mt	Megaton
CBC	Cost Benefit Calculations	NEMO	Next Energy Modeling system for Optimization
COP	Conferences of Parties	LCPDP	Least Cost Power Development Plan
C&I	Commercial & Industrial	LEAP	Low Emissions Analysis Platform
DERs	Distributed Energy Resources	KPLC	Kenya Power and Lighting Company
EPRA	Energy and Petroleum Regulatory Authority	O&M	Operating and Maintenance
EP	Energy Productivity	OSeMOSYS	Open-Source Energy Modeling System
ESSs	Energy Storage Systems	PA	Paris Agreements
IT	Imports Tariff	PV	Photovoltaic
GDP	Gross Domestic Product	RE	Renewable energy
GHG	Greenhouse Gas	RERs	Renewable Energy Resources
GoK	Government of Kenya	SC	Small Commercial
GWP	Global Warming Potential	SEI	Stockholm Environmental Institute
MILP	Mixed Integer Linear Programming	SSA	Sub-Saharan African
MOE	Ministry of Energy	T&D	Transmission & Distribution
		TFCD	Task Force on Climate-related Financial Disclosures
		WHO	World Health Organization

result, energy storage systems (ESSs) complement these aforementioned shortfalls [7].

This study aims to develop alternative, more sustainable electricity development pathways and investigate their impacts on system costs and greenhouse gas emissions in Kenya compared to the existing LCPDP 2017–2037. The scenarios explored include deep penetration of RERs along with energy storage systems, sustainable policy instruments to drive productivity, and internalizing of environmental externalities.

Energy planning models (EPMs), are key tools for energy planners and decision-makers to evaluate energy development and analyze greenhouse gas emissions while considering different technological and policy interventions. LEAP modeling software with NEMO (Cbc) optimization framework is used to model and simulate the different pathways and analyze the impacts of GHGs and system costs.

Following a review of the existent literature, it could be observed that most of the energy planning models do not consider the energy storage systems (ESSs) to enhance the reliability of future power systems. Moreover, some existing scholarships fail to simultaneously investigate the socio-techno-economic impacts associated with energy saving as well as the reduction of greenhouse gas emissions. The internalization of environmental externalities and energy productivity has also not been taken into consideration in most energy planning models. The present study fills this knowledge gap and contributes to the literature by considering energy storage systems in the energy planning

scenarios, including the analysis of socio-techno-economic impacts of electricity development pathways on system costs and greenhouse gas emissions.

2. Theoretical background

2.1. Energy scenario development

Energy scenario development has emerged as an important tool enabling energy analysts and planners to explore the impact of climate-related risks and opportunities. Scenarios have been used to explore the impact of the unchanging state of affairs, different policy instruments, alternative pathways for decarbonization, and different mitigation strategies on energy usage and GHG emissions. The Task Force on Climate-related Financial Disclosures (TFCD) recognizes the use of scenarios in analyzing climate-related issues and their potential financial implications aiding in decision-making for investors and risk management teams [8].

2.2. LEAP/NEMO platform

The model of the electricity mix of Kenya is built based on data collected from available literature such as LCPDP 2017–2037, Economic survey 2020, Vision 2030 projects, Kenya Power annual reports, and

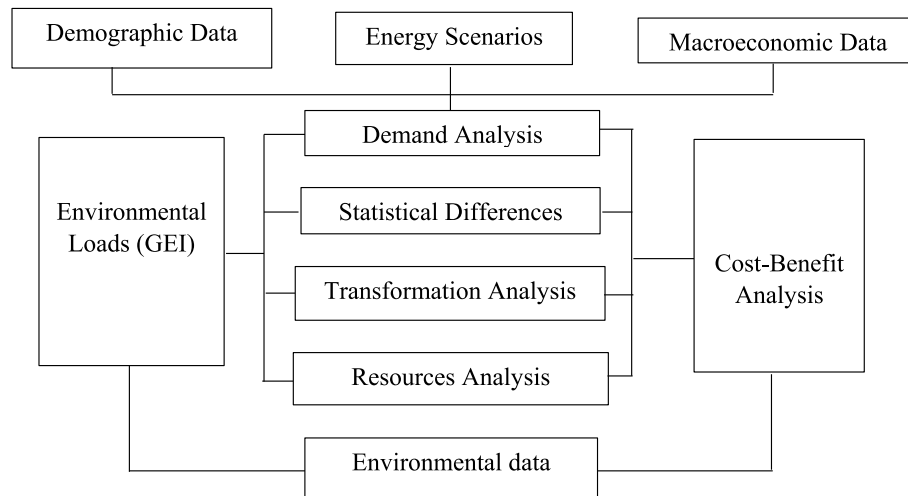


Fig. 1. Low emission analysis platform framework diagram.

Table 1
Vision 2030 flagship projects.

Project	Reference				High			
	First year of operation	Initial load [MW]	Year of total load	Total load [MW]	First year of operation	Initial load [MW]	Year of total load	Total load [MW]
Electrified mass rapid transit system for Nairobi	2024	15	2030	50	2022	15	2027	50
Electrified gauge standard railway Mombasa-Nairobi	2022	98	2030	130	2021	100	2028	300
Electrified gauge standard railway Nairobi-Malabar	2026	61.74	2035	61.74	2024	63	2032	189
Electrified LAPSET gauge standard railway	–	–	–	–	2035	30	2037	30
Oil pipeline and Port Terminal (LAPSET)	2025	50	2037	150	2022	50	2032	150
Refinery and Petrochemical Industries (LAPSET)	2028	25	2037	100	2025	50	2030	200
Konza Techno city	2024	2	2037	190	2022	2	2034	200
Special Economic Zones	2021	5	2037	110	2020	30	2028	110
Integrated steel mill					2030	100	2035	200

Kenya power system curve, among others, as well as some assumptions which are discussed in subsection 3.5. The modeling tools that have been used are the Low Emissions Analysis Platform (LEAP) and the Next Energy Modeling system for Optimization (NEMO). LEAP software [9] is a well-applied software for energy planning, scenario analysis, and climate change mitigation. It has been used by students, governments, non-governmental organizations, and private companies for research, energy planning, and GHG abatement strategy formulation as well as a basis for some nations' intended nationally determined contributions. LEAP is a bottom-up model which can integrate all sectors of an economy and calculate GHG emissions for both energy and non-energy sectors. LEAP can be paired with either the Open-Source Energy Modeling System (OSeMOSYS) or the Next Energy Modeling system for Optimization (NEMO) based on the needs of the users. NEMO is particularly useful over OSeMOSYS due to its ability to model energy storage. The mixture of LEAP and NEMO exhibits outstanding optimization features for energy planners and analysts due to NEMO's powerful optimization abilities [9]. The important features of NEMO are that it enables the least-cost optimization of energy supply, modeling of emissions constraints and renewable energy targets as well as energy storage modeling [10]. NEMO can indeed be used with many types of solvers. Fig. 1 depicts the LEAP framework diagram [9].

2.3. Kenya's LCPDP 2017–2037

Kenya's LCPDP 2017–2037 is a Kenya Energy Sector Report intended to guide the energy sector on generation expansion opportunities, transmission infrastructure target network expansion as well as resource

requirements for the expansion program over a long term of 20 years. It is an update of the 2015–2035 electricity master plan taking into consideration the Feed in Tariff policy and focusing on the Big 4 Agenda program in which energy is expected to be a key pillar of the program. On the one hand, the Ministry of Energy (MOE) sets the policy guidelines and on the other hand, the regulatory requirements are provided by the Energy and Petroleum Regulatory Authority (EPRA). The report covers a comprehensive load forecast and addresses the committed generation projects between 2017 and 2024 and also the expansion program for the period 2025–2037 [11].

2.4. Data requirements

Data required for long-term energy planning includes but is not limited to demographic, fuel use, energy intensity, historical energy generation and demand, existing and future planned power plant specifications, transmission and distribution data, resource specifications, emission factors, costs, interest, and inflation rates among others. According to the 2019 census [12], urban and rural portions of the population in Kenya are at 31.2% and 68.8%, respectively. The electrification rate is 84% and 71.7% in urban and rural households, respectively. The urban saturation of lighting fuel used is 99% electricity, 16.5% kerosene, and 19.3% solar. For rural households, the saturation of lighting fuel used is 28% electricity, 16.5% kerosene, and 19.3% solar [12].

Moreover, the transmission and distribution losses negatively influence the power system. Those are expected to drop to 16.6% in the end year 2037 from 23.7% in the base year 2019. The transmission & distribution (T&D) losses increased to 23.7% in 2019 from 21% the

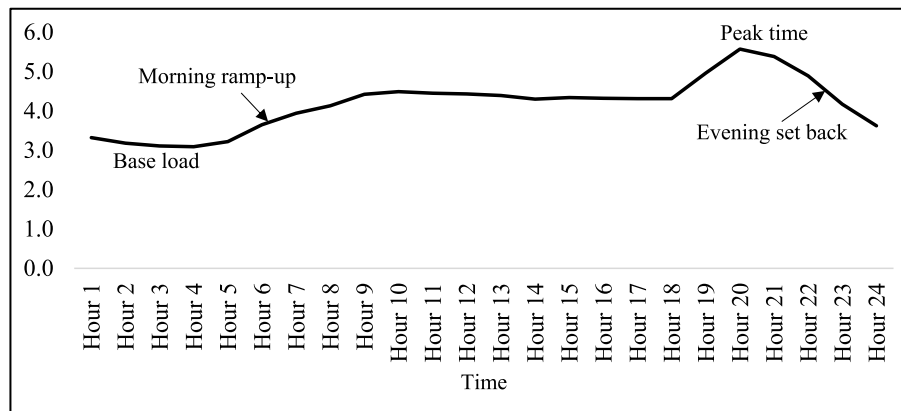


Fig. 2. Energy Load Shape (% of annual load) for Kenya in 2019.

previous year owing to transmission system constraints, growth in the distribution network, and an increase in commercial losses, that adversely affected the financial performance of the power system [13]. It can be deduced that there is an annual average energy loss reduction of 0.4%.

The projects related to Kenya's vision 2030 agenda are expected to greatly affect the electricity development trajectory in the country. Those projects are presented in Table 1 [11]. The reference scenario presents the updated LCPDP 2019–2037 while the high penetration RERs prioritizes renewable energy resources.

The energy demand for each project per year, post commissioning can be forecasted using the formula:

$$VC = \sum TL(MW) \times Lf \times 8760 \text{ hrs} \quad (1)$$

where VC is the vision consumption, TL denotes total load, and LF is the load factor.

$$\text{Capacity Before Additions} = (\text{Exogenous Capacity} + \text{Endogenous Capacity Added Previously}) * \text{Capacity Value} \quad (5)$$

The energy system load shape for Kenya over 24 h is depicted in Fig. 2.

As shown in Fig. 2, the peak demand occurs in the evening between 18h and 23 h. This data is particularly useful to plan for the required reserve margin.

Energy productivity is the ratio of economic output per unit of energy use. It can inform governments about their relative performance on interrelated and often conflicting objectives related to economic parameters, energy, and environmental issues [14].

$$\text{Reserve Margin Before Additions} = \frac{(\text{Capacity Before Additions} - \text{Peak Requirement})}{\text{Peak Requirement}} \quad (7)$$

Energy productivity is given such that;

$$\text{Energy productivity} = \frac{GDP}{\text{Electricity consumption}} \quad (2)$$

$$\text{Endogenous Capacity Additions Required} = (\text{Planning Reserve Margin} - \text{Reserve Margin Before Additions}) \times \text{Peak Requirement} \quad (8)$$

where GDP represents the Gross Domestic Product.

Besides, the estimated damage costs from CO₂ emissions are EUR 33,6 per ton reflecting the marginal costs of abatement [15].

2.5. Scenario formulations

The concept of scenario analysis is the core of LEAP depicting how future energy systems might expand over time in a particular socioeconomic and under a set of policy conditions. The LEAP scenario formulations are defined in [16].

The LEAP calculates the final energy demand analysis as follows:

$$D_{b,s,t} = TA_{b,t,s} \times EI_{b,t,s} \quad (3)$$

where D denotes energy demand, TA represents total activity, EI denotes energy intensity, b represents the branch, s is scenario and t denotes the year.

The total activity level for a technology is the product of the activity levels in all branches from the technology branch back up to the original demand branch; such that;

$$TA_{b,t,s} = A_{b',t,s} \times A_{b'',t,s} \times A_{b''',t,s} \times \dots \quad (4)$$

where A_b denotes the activity level in a particular branch b , b' is the parent of branch b , b'' is the grandparent, etc.

From the transformation calculations perspective, LEAP evaluates the capacity added as follows:

Existing capacity before the addition of endogenously calculated additions is given by Equation (5).

Peak system power requirements on the module are calculated as a function of the total energy requirements and the system load factor as in Equation (6).

$$\text{Peak Requirement}[MW] = \frac{\text{Energy Requirement}[MW - hr]}{\text{Module Load Factor} * 8760[\text{hrs/year}]} \quad (6)$$

The reserve margin before the addition is formulated as:

The amount of endogenous capacity additions required is written as follows:

From the process dispatch calculations viewpoint, LEAP software dispatches the processes as in Equation (9).

Process share in the proportion to base year output is formulated as

$$\text{ProcessShare}_i = \frac{\text{Base year output}}{\sum_{i=1 \dots n} \text{Base year output}} \quad (9)$$

The available Capacity is given by

$$\text{ProcessShare}_i = \frac{\text{Capacity}_i \times \text{MCF}_i}{\sum_{i=1 \dots n} \text{Capacity}_i \times \text{MCF}_i} \quad (10)$$

In terms of dispatching processes of a load curve, LEAP calculates the dispatch by running cost as follows:

$$\text{Running cost}_i = \text{VariableOMCost}_i + \text{FuelCost}_i / \text{Efficiency}_i \quad (11)$$

Further, the peak system power requirement is formulated as follows [17]:

$$\text{Peak System Power Requirement} = \frac{\text{Energy Requirement [MWh]}}{\text{Load Factor} \times 8760 [\text{h/year}]} \quad (12)$$

2.6. Energy planning pathways studies

A lot of research has been carried out on energy planning pathways. For instance, the authors in Ref. [18], examined the extent to which large-scale renewable energy resource is required to manage Kenya's greenhouse gas emissions based on a business as usual (BAU) scenario. However, this study did not consider the intermittent and stochastic nature of renewable energy technologies (RETs). In Ref. [19], three possible power development scenarios related to cost and GHG emissions with LCPDP as a reference scenario were investigated. The optimization approach and the socio-economic investigation were not considered in this study. In Ref. [20], a non-parametric energy planning for assessing long-term energy policy scenarios in Greece was examined. In Ref. [21], the authors analyzed the optimal generation scenarios for sustainable power generation in Ghana using OSeMOSYS. Although the optimization strategy was examined, their LEAP model did not consider ESSs. In Ref. [22], the authors analyzed the energy demand and supply by considering an economic scenario such as GDP as well as the national policy for energy efficiency in Ecuador. In Ref. [23], the authors examined the surplus of electrical energy produced and suggested reducing the total energy cost in rural regions of Nepal. In Ref. [24], the authors investigated long-term electrical energy demand forecast and supply for Pakistan using LEAP for policy analysis. However, the forecasted LEAP model framework did not consider the battery storage systems. In Ref. [25], the authors analyzed urban energy planning and carbon mitigation policies for Beijing city in China under three scenarios; BAU, basic policy, and low carbon. The social development of the region was not tackled. In Ref. [26], a forecast electricity demand-supply for Maharashtra in India considering BAU, energy conversion, and renewable energy scenarios. However, this study did not take into consideration GHG emissions mitigation. In Ref. [27], optimal energy transition pathways applied to Italy for long-term planning were investigated using EnergyPLAN software. The CO₂ emissions abatement was also considered in the formulated cost-optimization problem. In Ref. [28], an optimization energy planning model was analyzed using a stochastic mixed integer linear programming model (MILP) considering the uncertainty of the Greek power system. In Ref. [29], the authors presented an integrated rural energy planning framework for sustainable energy development in developing countries. In Ref. [30], future energy demand-supply and associated GHG emissions were explored for Nigeria's power system. Moksnes et al. [31], investigated the energy pathways development strategies for Kenya allowing the country to reach its electrification goal for medium to long-term energy planning. Yang et al. [32] examined critical transformation pathways and socio-environmental benefits of energy transition using LEAP. Nikolaos et al. [33] developed a spatial multi-period long-term energy planning model for the Greek power system. Hu et al. [34] developed sustainable urban energy planning aiming at combining energy planning with economic analysis while reducing energy consumption at a minimal cost, particularly in Shenzhen China. Emodi et al. [35] explore the least-cost electricity generation expansion options while examining the effectiveness of emission reduction policies under climate change. Felver [36] modeled GHG emissions for Azerbaijan until 2030 under a business as usual scenario, a scenario with currently planned climate change mitigation measures, and a final scenario mirroring EU policy plans. Hernandez et al. [37] examined the estimated industrial emissions for a

Table 2

KPLC sales by customer category in GWh.

Customer Category	2014/ 15	2015/ 16	2016/ 17	2017/ 18	2018/ 19
Domestic – DC	1866	2007	2138	2335	2366
Small Commercial – SC	1143	1153	1201	1222	1250
Commercial & Industrial – CI	4030	4104	4266	4225	4462
Street lighting – SI	35	40	55	66	68
TOTAL	7090	7330	7701	7881	8147
% Increase P.A.	5.0%	3.4%	5.01%	2.3%	3.4%

long-term period in Columbia. Dobravec et al. [38], analyzed a multi-level governance energy planning and planning by focusing on the alignment between the local energy and climate initiatives and the national and EU goals. In Ref. [39], static and artificial neural network models were analyzed for electricity demand forecasting in the Brazilian industrial sector. The results indicated that the models can adjust the data, capture the complex structure and present suitable predictive energy consumption. In Ref. [40] the authors investigated long-term energy planning considering heating and electricity consumption using deep learning. As it can be seen from literature, energy planning models and energy policy are very significant approaches to elaborate and plan for long-term electrical energy infrastructure to cope with uncertain future energy demand-supply as well as tackling the main challenge of climate change and GHG reduction. There have been different types of scenario schemes for energy planning in literature. Each approach of energy planning model presents its merits and demerits. However, most of the energy planning scenarios models did not consider the energy storage systems (ESSs) to enhance the reliability of future power systems owing to the intermittent nature of renewable energy sources. Additionally, some research fails to simultaneously investigate the socio-techno-economic impacts associated with energy saving as well as the reduction of greenhouse gas emissions. Furthermore, the internalization of environmental externalities and energy productivity has not been taken into consideration for most energy planning models. Consequently, this paper analyzes the socio-techno-economic impacts of electricity development pathways on system costs and greenhouse gas emissions considering energy storage systems. Most importantly, three scenarios are considered: high penetration of RERs, internalizing of environmental externalities, and energy productivity.

3. Methodology

To evaluate the future in terms of energy planning pathways, environmental effects, and socio-economic impacts, an integrated computer-based energy-environment modeling tool for assessing energy policies and sustainable energy plans model termed Low Emissions Analysis Platform (LEAP) with Next Energy Modeling system for Optimization (NEMO) is used for Kenya's energy sector. NEMO is a high-performance, open-source energy system optimization modeling tool. The fundamental benefits of NEMO include least-cost optimization of energy supply and demand, support for multiple regions and regional trade, modeling of energy storage, modeling of emissions and emission constraints (including carbon prices and pollutant externalities), modeling of renewable energy targets, numerous performance tuning options, among other benefits [16]. Therefore, NEMO presents a better opportunity for achieving the objectives of this study than the other models.

Different customer category sales are often clustered in the electricity market by the local utility company, Kenya Power and Lighting Company (KPLC). The sales by customer category from 2014 to 2019 apart from the IT tariff category that no longer exists under the current tariff structure is shown in Table 2 [41].

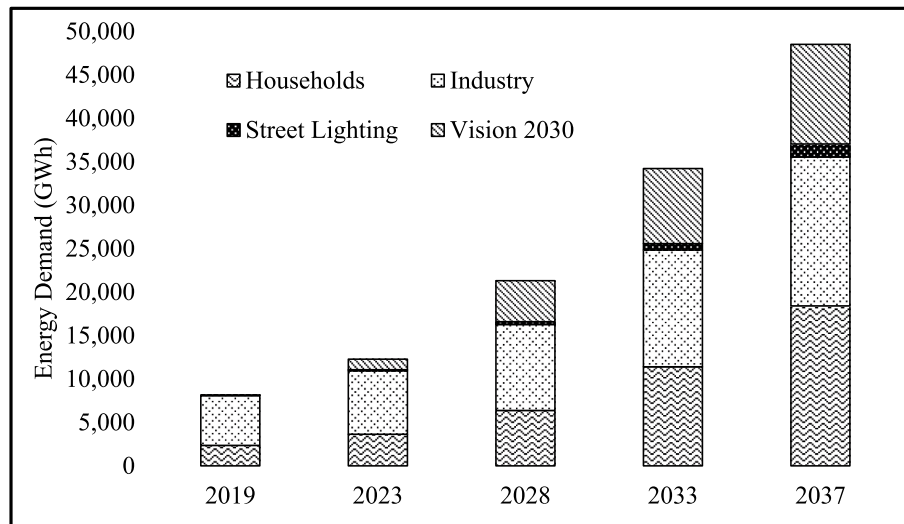


Fig. 3. Energy demand projections 2019–2037.

3.1. Model input data collection and sources

Input parameters are collected from; LCPDP 2017–2037, Economic survey 2020, 2019 Kenya Population and housing census results, Vision 2030 projects, Kenya Power, Kenya power system curve), World bank

data, KenGen Power Plant Data 2019 and Cost data from Energy Information Administration 2020, Global wind atlas, and solar GIS maps among others. In this study, the cost of CO₂ is taken from industrial facilities in Europe [15].

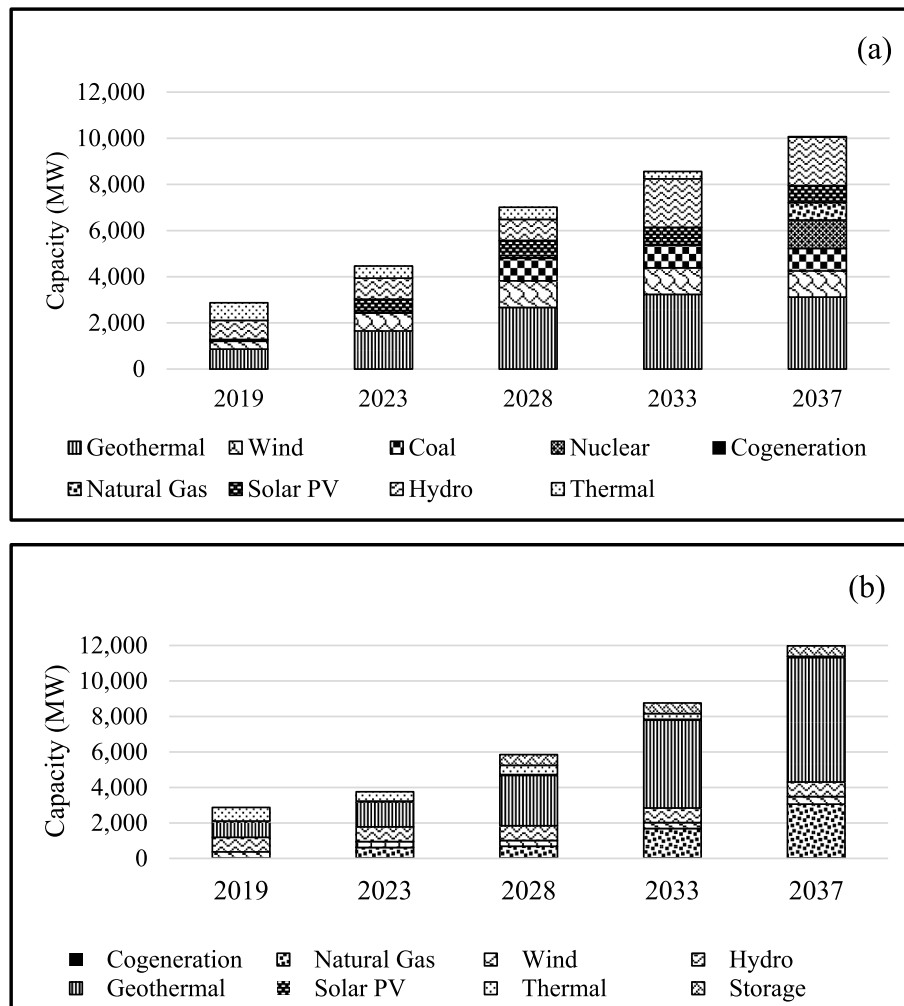


Fig. 4. Capacity of power plants; (a) LCPDP 2019–2037 and, (b) High penetration of RERs with energy storage.

Table 3

Capacity of power plants under LCPDP and high renewables with storage for the end year 2037.

Resources	% Plant Capacity in the year 2037	
	LCPDP	High RERs with ESS
Geothermal	31	58
Hydro	21	7
Nuclear	12	–
Wind	11	4
Coal	10	–
Solar PV	7	0.4
Natural gas	7	25
Cogeneration	0.3	–
Thermal	0.2	0.15
Storage		5

3.2. Model setup in LEAP

The Kenya LEAP Model is modeled with 2019 as the base year upon which future scenarios are analyzed. Energy demand data is classified as households, industrial, street lighting, and Vision 2030 projects. The emission factors for each fuel type were applied both at the point of use and at generation where applicable. Electricity generation processes and plant specifications, transmission and distribution losses, and resource data are structured in their respective modules. The costs of power

plants and fuel are also specified. For the reference scenario, the gross electricity consumption grows from 10,465 GWh in 2017 to 14,334 GWh and 39,187 GWh in 2022 and 2037, respectively as per Table 1. This represents an average annual growth of 6.7% per annum [11]. In Table 2, the electricity sale grows by 3.4% [41].

3.3. Scenario simulations

The reference scenario is created by accounting for demographic changes from 2019 to 2037, energy intensity changes, committed and planned expansion projects as well as planned decommissioning. The high RERs with energy storage scenario on the other hand considers resource availability, reserves, and yields as well as energy storage specifications and are optimized using NEMO Cost-benefit calculations (CBC). For internalizing environmental externalities carbon dioxide, nitrogen dioxide, and sulphur dioxide were charged per unit of emission whereas for the energy productivity scenario, energy intensity data, and reduction in transmission and distribution losses were explored.

The analysis view in LEAP can be summarized as i) Create the data structures for the selected sites in terms of key variables, demand, transformation, and resources; ii) Create and manage alternative scenarios using the manage scenarios screen; iii) Enter the data, assumptions and modeling relationships for each scenario using the data entry tables on the right-hand side of the screen.

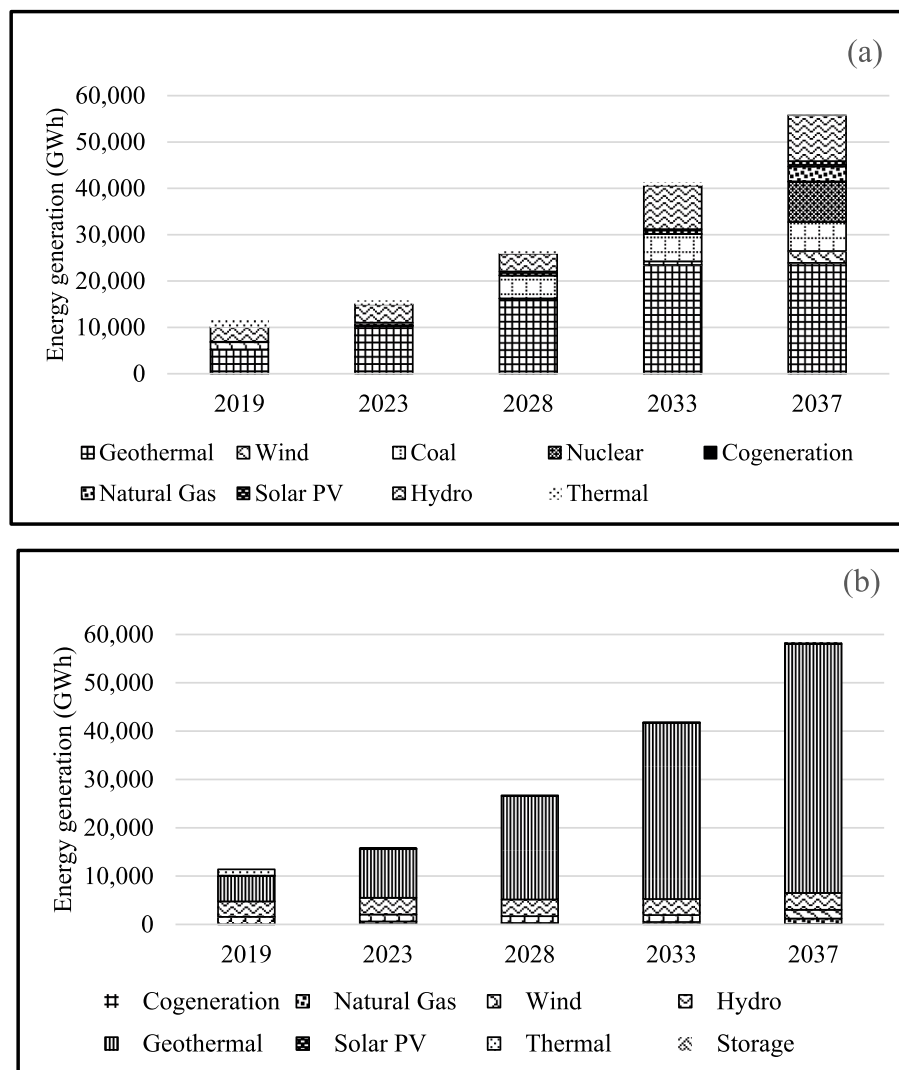


Fig. 5. Energy generation; (a) LCPDP 2019–2037 and, (b) High penetration of RERs with Storage.

Table 4

Energy generation under LCPDP and high RERs with ESS for the end year 2037.

Resources	% Energy generation	
	LCPDP	High RERs with ESS
Geothermal	55	81
Wind	3	5
Coal	13	–
Nuclear	2	–
Cogeneration	0.1	0.1
Natural gas	2	1.4
Solar PV	3	0.5
Hydro	20	12
Thermal	2	0.3

3.4. Economic analysis procedures

The economic analysis exhibits the cost-benefit of the various scenarios. In this research, the cost-benefits are broadened by exploring environmental externalities as well as the capital, fixed, and variable O&M costs for the different power plants. The resource analysis screens in LEAP are used to enter data for power generation plants, including both renewable and non-renewable energy.

3.5. Assumptions

To carry out this research, the projections for street lighting are based on previous annual growth rates and extrapolated linearly at 18%. In other sectors the growth assumptions are made in the line with LCPDP as follows: the population growth is estimated to be 2.5%. Urbanization growth at 5.81%, industrial energy intensity growth at 6%, and household energy intensity growth at 6%.

3.6. Model validation

The LEAP model has been validated with 2016 electricity generation data. The year 2016 is selected because of the availability of CO₂ emissions data for the Kenyan grid. The CO₂ equivalent emissions in 2016 generated from electricity in Kenya was 116 g of CO₂ per Kilowatt hour [42]. In addition, the electricity generation was 9971 GWh which includes transmission and distribution losses [12]. From this viewpoint, it follows that the total emissions are 1.156 Mt CO₂ equivalent. Applying the energy generation data in LEAP, the model yields a result of 1.207 Mt CO₂ equivalent. This implies a variance of 4.4% showing that the model presents a good representation of the real situation and, that it is effectively suitable for the Kenyan electric power system.

4. Results and discussions

4.1. Energy projections based on LCPDP

In this study, the forecasted Kenya's LCPDP 2019–2037 is used as the reference scenario. The useful energy demand of the key sectors namely households, industry, and street lighting as well as the vision 2030 projects is depicted in Fig. 3.

As shown, households exhibit the greatest energy demand growth rate due to the demographic and economic changes anticipated followed by the industrial sector. At the end year 2037, the energy demand of households and industries are 38% and 35%, respectively, while vision 2030 projects account for 24% and street lighting is 3%. Total useful energy demand (excluding transmission and distribution losses) is projected to be 48,514 GWh in 2037.

4.2. Projected plant capacity and energy generation mix under high penetration of RERs with ESS

The power plant capacity mix is presented in Fig. 4. Fig. 4 (a) shows the projections for LCPDP plant capacity accounting for the planned decommissioning of the thermal plants and the planned addition of the coal, nuclear, and natural gas plants. The system capacity mainly comprises geothermal, hydro, and wind as RERs. The geothermal power plant represents the greatest capacity power plant in the LCPDP development pathway strategy in Kenya.

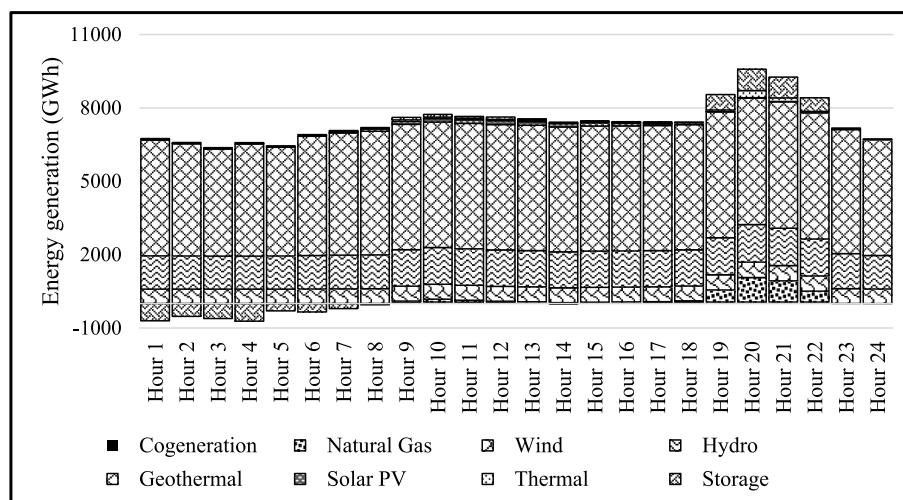
Fig. 4 (b) shows the capacity of power plants considering the high penetration of RERs with storage optimized by the LEAP-NEMO (CBC) optimization framework. The system mainly comprises geothermal, natural gas, and hydro. The planned capacity in high RERs with storage scenario is 1900 MW more than the LCPDP scenario accounting for the stochastic nature of the renewable resources which results in lower resource availability, capacity factors, and reduced ability to contribute to peak power requirements.

Table 3 further presents a comparison of actual shares of the energy mix in the development pathway strategies for the year 2037.

As it can be observed in Table 3, the major source of renewable energy electricity at the end of the year 2037 in Kenya comes from geothermal energy.

Energy generation for the two scenarios is further depicted in Fig. 5 with Fig. 5 (a) and (b) showing the trend under LCPDP and RERs with ESS, respectively.

In the LCPDP, the energy from RERs supplies 80.5% of the overall energy demand by the end year 2037. The breakdown of energy generation by resource type in LCPDP scenario and under high RERs with

**Fig. 6.** Hourly energy generation on a 24-h time slice, year 2028.

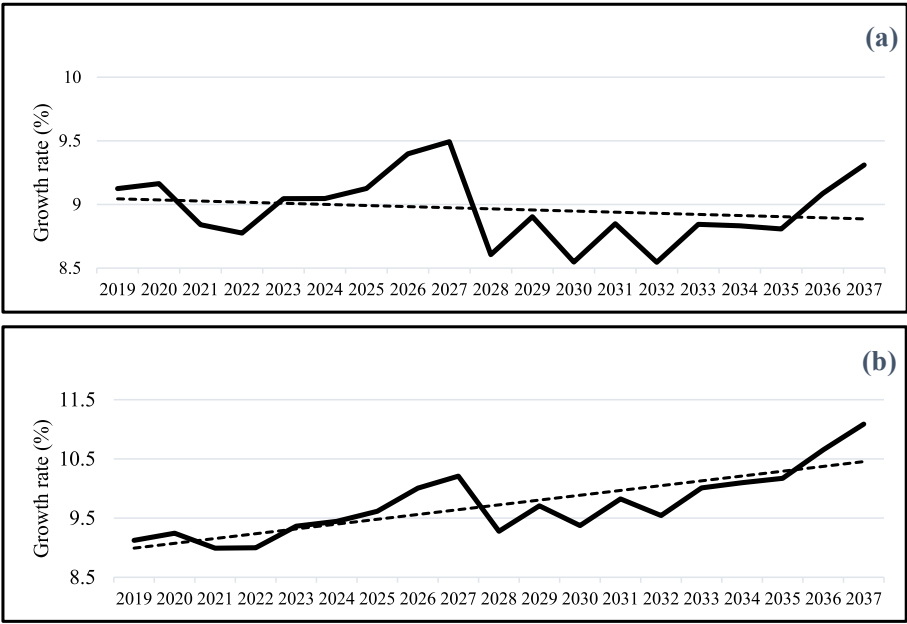


Fig. 7. Energy productivity; (a) LCPDP 2019–2037 and, (b) with two policy instruments.

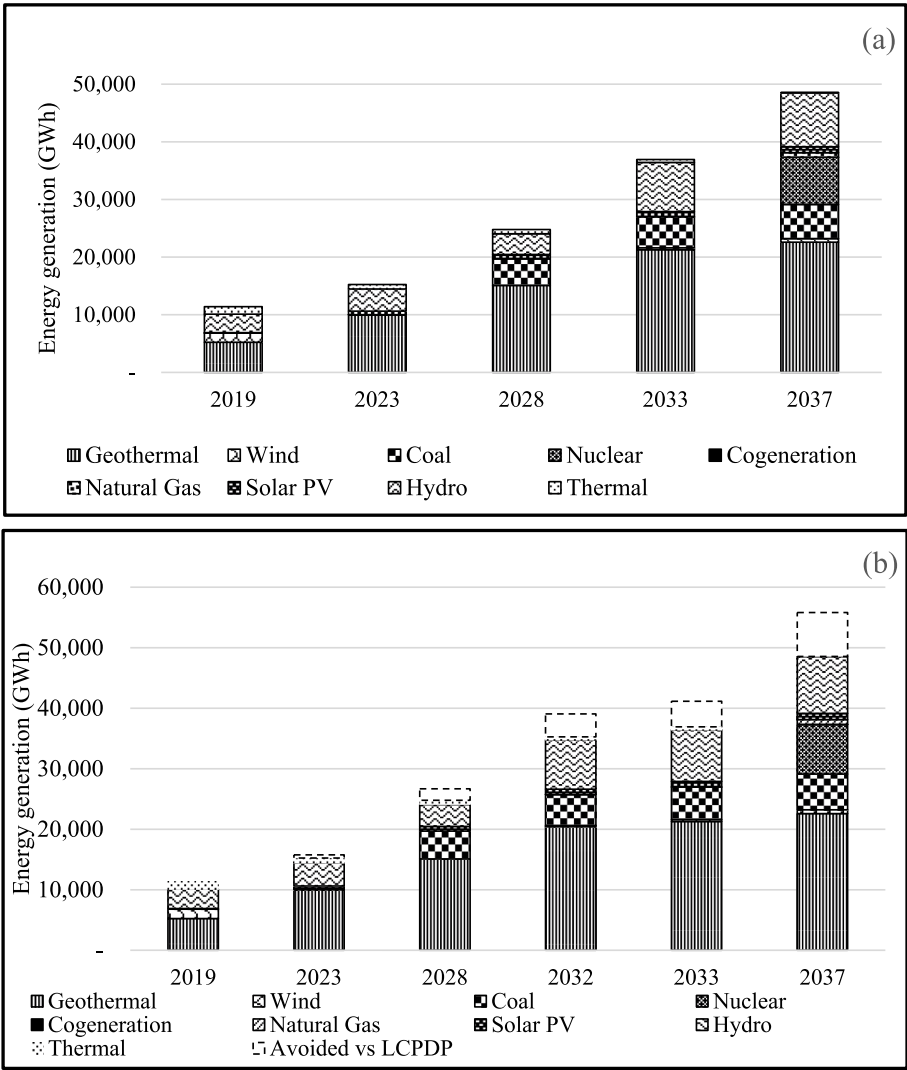


Fig. 8. Energy productivity scenario energy generation (a) LCPDP 2019–2037 and, (b) the avoided EP vs LCPDP.

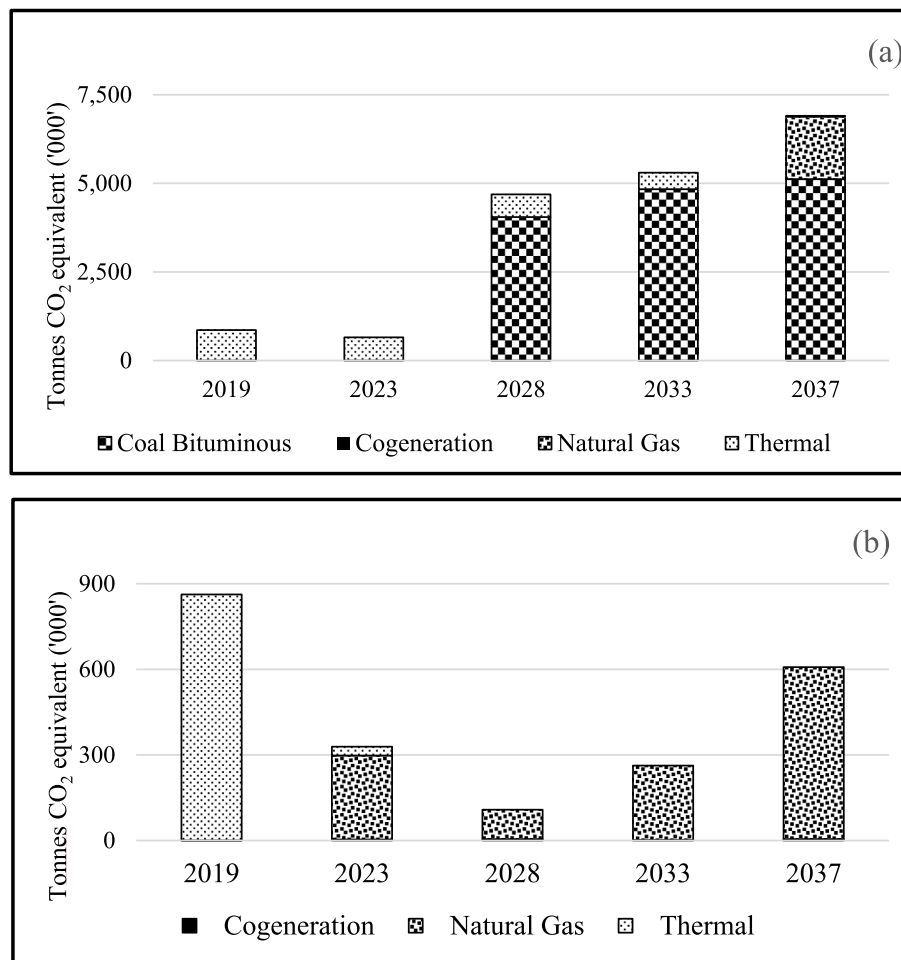


Fig. 9. 20 years GWP; (a) LCPDP 2019–2037 and, (b) high penetration of RERs with Energy Storage system.

ESS for the end year 2037 are represented in Table 4.

To meet the rapid demand of energy consumption and increase the reliability of power systems, the high incorporation of RERs with energy storage system plays a vital role in future energy planning of power system pathways. Table 4 indicates the energy generation considering the RERs along with the ESS. As shown in Fig. 5 (b) and Table 4, the high penetration of renewable energy resources yields 98.4% energy generation from renewable energy resources. It is worth noting that geothermal energy resource provides the greatest energy source approximately 81%, followed by hydro energy at 12%, wind energy at 5%, solar PV at 0.5%, and cogeneration at 0.2%. The non-renewable energy that consists of thermal and natural gas provides 0.3%, and 1%, respectively. Solar PV's low installation capacity is attributed to its low resource availability, low process efficiency, and the fact that it is not considered during peak power requirements. Solar PV hence remains a better candidate for decentralized power generation. Ghana is a country that is comparable to Kenya in terms of GDP per capita and renewable energy resources. An investigative study conducted for Ghana showed that renewable energy technologies can contribute more than 70% of its generation needs by 2040 [43], this further verifies the relevance of these results for the sub-Saharan region.

To observe the contribution of energy storage in the alternative scenario as earlier presented in Fig. 5 (b), the hourly energy generation is shown in Fig. 6.

The battery banks are charged from hour 1 to 7 and slightly at hour 14 using excess energy from the grid. The energy stored is dispatched from hour 9 to 13 and from hour 19 to 22 which is also the peak period of the Kenya power system curve (Fig. 2). Energy storage contributes 884

GWh which accounts for 3% of the projected energy requirements for 2028.

4.3. Energy productivity

Fig. 7 represents the energy productivity case. The EP of LCPDP 2019–2037 with no existing policies is shown in Fig. 7 (a). Further, the energy productivity with two policy instruments namely energy efficiency in households and reduction of transmission and distribution losses is depicted in Fig. 7 (b).

In Fig. 7 (a), it is observed that the energy productivity exhibits a downward trend due to high energy demand growth relative to GDP growth in the LCPDP reference scenario. However, exploring policy instruments that would drive energy efficiency up in domestic households by lowering the energy intensity growth rate to 3% instead of 6% as in LCPDP, and further employing another instrument that would further drive down the transmission and distribution losses to 15% from 16.6% by the year 2037, the energy productivity takes an increasing trend as shown in Fig. 7 (b). The household sector was selected as it exhibits the highest growth in energy demand in the said period.

The fluctuations observed in the years 2028, 2030 and 2032 are attributed to high load from Vision 2030 projects operationalization namely; Electrified standard gauge railway Mombasa-Nairobi, Refinery, and petrochemical industries (LAPSSET), electrified standard gauge railway Nairobi-Malaba and oil pipeline and Port Terminal (LAPSSET).

The assessment of energy generation for EP is presented in Fig. 8. Fig. 8 (a) presents the energy generation showing the potential impact of the two policy instruments such as the high potential of energy

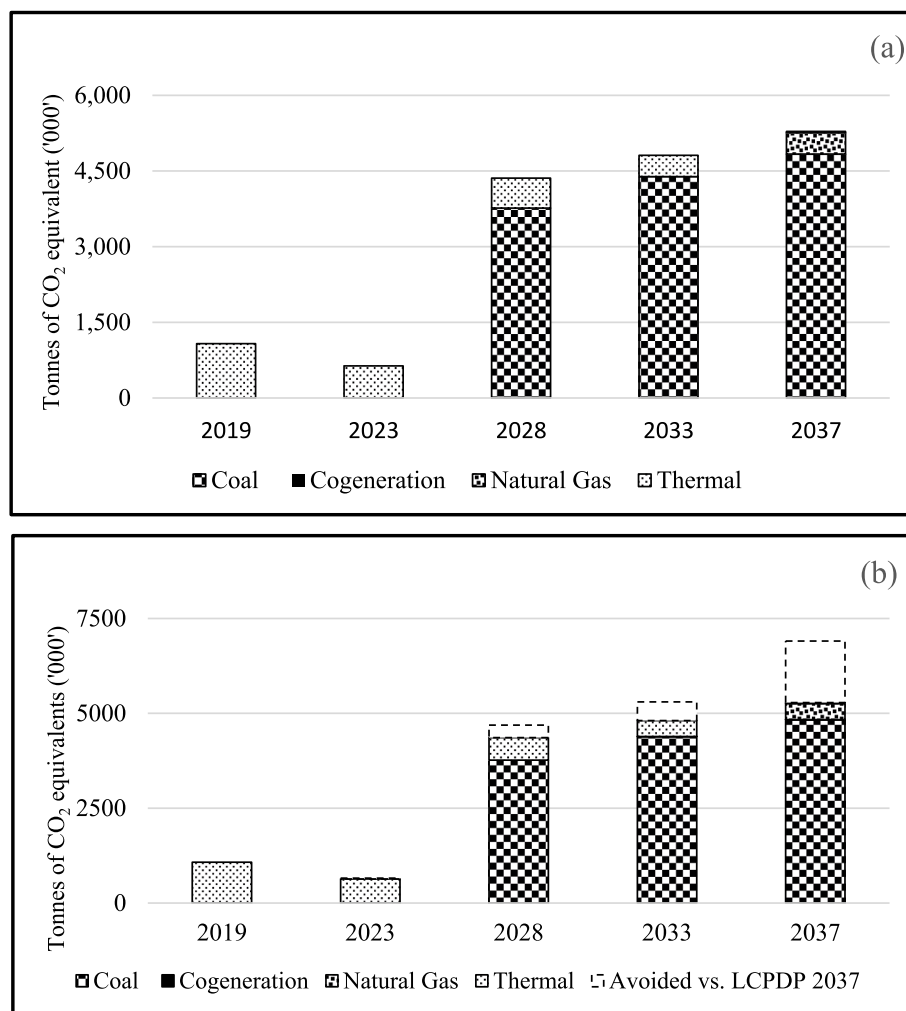


Fig. 10. 20 years GWP in energy productivity scenario; LCPDP 2019–2037 and, (b) emissions avoided versus LCPDP.

efficiency in households and reduction of losses at transmission and distribution level. The assessment of the energy productivity scenario in comparison with the LCPDP is shown in Fig. 8 (b).

The evaluation of the avoided energy while considering the energy productivity versus the LCPDP scenario is evaluated in Fig. 8 (b). The avoided energy generation at the end year 2037 is up to 7267.9 GWh which is a 13% decline in comparison to the LCPDP scenario.

4.4. GHG emissions

The GHG emissions in energy development pathways should be minimized to lower the negative effect of climate change. From the GHG emissions standpoint, the results of a 20-year Global Warming Potential (GWP) portray the warming effect that the generated greenhouse gases would yield in the atmosphere over 20 years. Fig. 9 (a) depicts the respective energy absorbed over 20 years for the reference scenario, LCPDP 2019–2037. The assessment of warming effects for the high penetration of RERs with energy storage system is then presented in Fig. 9 (b).

From Fig. 9 (a), coal exhibits the greatest quantity of emissions accounting for approximately 74%, natural gas emits 24%, while thermal and cogeneration emit 0.4% and 0.1%, respectively. In Fig. 9 (b), there is a significant reduction in emissions by around 30% as compared to the base year 2019. When further compared to the LCPDP energy planning scheme, the high penetration of renewable energy with storage yields a GHG emission reduction of 6.3 Mt of CO₂ equivalent representing a 91%

decline at the end year 2037. This is a great achievement in terms of the Paris Agreement that seeks a reduction of 100% the GHG by 2050. The low emissions in the middle years of high RERs with energy storage scenario are attributed to the transition through the installation of energy storage in 2025 which provides alternative power in the peak time reducing energy generation by natural gas which is the major polluter in this scenario.

The 20 years GWP is further analyzed in the energy productivity scenario and obtained results are depicted in Fig. 10 (a). For comparative reasons, the same is carried out in comparison with the LCPDP case in Fig. 10 (b).

It can be observed that the avoided emissions are up to 1.62 Mt CO₂ equivalent at the end year 2037 which represents a 24% decline compared to the LCPDP.

4.5. Economic analysis

From the cost viewpoint, all the scenarios are investigated to determine the most cost-effective energy planning pathways. The overall cost of the energy system in LCPDP includes the capital investment costs of each power plant, the fixed and variable operation, and maintenance (O&M) costs are presented in Fig. 11. Fig. 11(a) shows the cost for the reference LCPDP case. However, for a more objective and realistic comparison, Fig. 11 (b) further presents the overall cost in the LCPDP case while internalizing environmental externalities. This was arrived at by applying a charge on carbon dioxide, nitrogen oxides, and sulphur

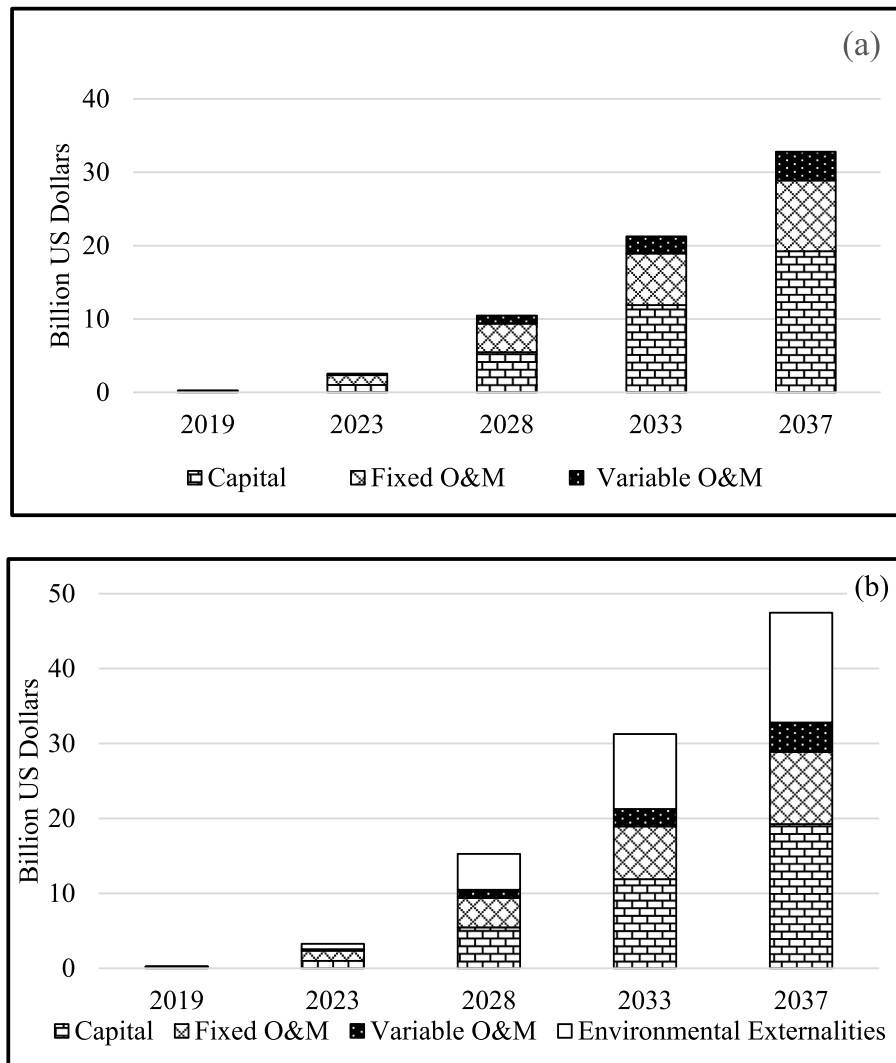


Fig. 11. Cumulative system cost; (a) LCPDP 2019–2037 and, (b) LCPDP considering internalizing environmental externalities.

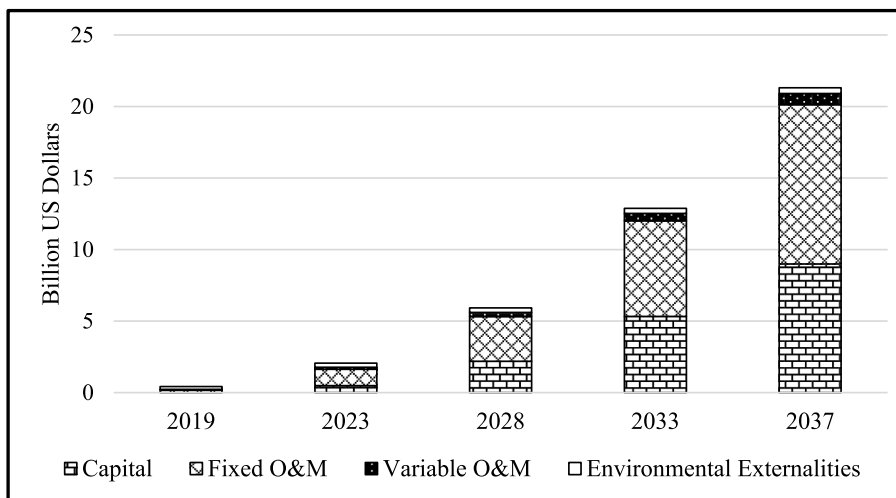


Fig. 12. Costs 98% RERs with storage considering internalizing environmental externalities.

dioxide emissions.

As shown in Fig. 11 (a), the cumulative system cost for LCPDP is 32.7 billion USD in the year 2037. The capital costs account for 59% of

overall system cost with the fixed and variable O&M at 29% and 12%, respectively. In Fig. 11(b), the overall costs are evaluated having internalized environmental externalities, which present an economic

concept of uncompensated environmental effects of production and consumption that affect health issues paid by consumer utilities and enterprises. This demonstrates the social costs associated with the choice of power production technology mix as exemplified in the LCPDP case. When environmental externalities are internalized and accounted for by use of tax on emissions, the overall cost of production increases to 47.5 billion USD. It then follows that 41% of the overall system cost is capital, 20% is fixed O&M, 8% is variable O&M costs and 31% is environmental externalities. There is a 45% increase equivalent to 14.8 billion USD compared to the overall system cost of the reference scenario LCPDP 2019–2037 which had not internalized the externalities.

Similarly, the assessment of the costs is analyzed in high penetration of RERs while internalizing environmental externalities as shown in Fig. 12.

By the year 2037, the total cost of the system is 21.3 billion USD which is a decrease of 35% (from USD 32.7 billion) and 55% (from USD 47.5 billion) compared to the LCPDP without and with internalized environmental externalities, respectively. In this scenario, the overall cost is distributed as, 52% of all costs are fixed O&M, 42% are capital costs, 4% variable O&M, and 2% of internalizing environmental externalities costs.

5. Conclusions

This paper proposes the optimized energy development pathway considering the high penetration of renewable energy resources and energy storage system. The reference scenario is based on the Kenyan government's power expansion plan LCPDP 2017–2037 and existing policies. The study is conducted in LEAP with NEMO optimization. The obtained results show that 98.4% of RERs and battery energy system is a feasible energy planning pathway. The combination of RERs and ESSs presents great benefits in terms of cost-effectiveness, CO₂ emissions reduction, reliability improvements in the future, and security of Kenya's power system. Consequently, the effectiveness of these proposed energy planning pathways has shown considerable advantages compared to the current LCPDP 2017–2037 energy development pathways. The cost savings are up to 55% which is equivalent to 26.1 billion US dollars and abates up to 6.3 Mt of CO₂ emissions at the end year 2037 which is a 91% reduction in comparison to the reference scenario. When environmental externalities are internalized and accounted for use of tax on emissions, the overall cost of production increases to 47.5 billion USD. It then follows that 41% of the overall system cost is capital, 20% is fixed O&M, 8% is variable O&M costs and 31% is environmental externalities. There is a 45% increase equivalent to 14.8 billion USD compared to the overall system cost of the reference scenario LCPDP 2019–2037 which had not internalized the externalities. In Kenya, geothermal energy resources present the major source of energy generation for the proposed energy planning pathways. This high penetration of RERs with energy storage has the potential to aid in the achievement of Kenya's nationally determined contributions as well as reduce climate change impacts on people and the planet. Policy instruments that drive energy efficiency in households, industries, and commercial facilities and that considerably lower transmission and distribution losses offer a promise to improve the operating performances of the power system.

Credit author contributions

Wambui V.: Conceptualization, methodology, investigation, formal analysis, validation, data curation, Writing-original draft, resources. Njoka F.: Visualization, Supervision, Writing – review & editing, project administration, resources. Muguthu J.: Supervision, Methodology, Writing – review & editing. Ndwalu P.: Writing – review & editing, resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors wish to express their gratitude to Charlie Heaps and LEAP/SEI for the provision of the LEAP modeling software; Mr. Haron Njiru for health data support; Dr. Xavier Musonye for availing useful data.

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